Deep Learning & Transformers Primer

1. Attention Is All You Need

The Transformer architecture introduced in 2017 discarded recurrent connections in favor of self-attention mechanisms. Self-attention allows tokens to compute relational weights against all other tokens in a sequence simultaneously in O(N^2) complexity.

2. Multi-Head Attention Mechanics

Queries (Q), Keys (K), and Values (V) are projected into multiple representation subspaces. The scaled dot-product attention formula is: Attention(Q, K, V) = softmax((Q * K^T) / sqrt(d_k)) * V. Scaling by the square root of dimension d_k prevents softmax gradients from vanishing in regions with extremely large values.

3. Pretraining and Fine-Tuning Paradigms

Modern LLMs undergo self-supervised pretraining on massive web-scale corpora, followed by Supervised Fine-Tuning (SFT) and Reinforcement Learning from Human Feedback (RLHF) or Direct Preference Optimization (DPO) to align model outputs with human intent.